Tiny but mighty parents: exploring insect parental care: a review

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Abstract
Parental care, a complex behaviour where parents invest in their offspring's survival and development is widespread in insects. Despite its importance, the evolutionary drivers, behavioural diversity and ecological consequences of parental care in insects remain poorly understood. This article synthesizes current knowledge on parental care in insects, highlighting its behavioural manifestations, forms of parental care, patterns of parental care, maternal and paternal behaviour among insects and adaptive significance. The role of environmental stress, mate selection and phylogenetic constraints in shaping parental care strategies. The ecological consequences of parental care, including its impact on offspring fitness and quality. Parental care in insects is a dynamic and multifaceted trait, influenced by a complex interplay of factors.

Keywords: Parental care, insects, evolution, behaviour and ecology.

Introduction
Any behaviour on the part of either male or female or both parent that enhances the chances of survival of offspring is termed as Parental care. Sometimes besides parents, other members of the family also help in taking care of the offspring. Parental care is traditionally defined as costly behavior by parents that increase the fitness of offspring, ranging from temporary egg-guarding to feeding of offspring by both parents up to adulthood. Care can be by females, males or both. Behavioral process of increasing the offspring survival prospects by protecting them from predators, to protect from food shortages, to prevent desiccation and to protect from a range of other environmental hazards. Parental care represents only one among several alternative solutions to overcome problems associated with environmental hazards that reduce offspring survival. Parents improve some aspect of offspring quality, which leads to an increase in offspring survival and or reproductive success in the future.

The evolution in parental care: The evolution in parental care of insects mainly dependent on the 3 factors

1. The harsh environmental conditions
2. Ephemeral or distant food sources and specialized foraging
3. Natural enemies

Types of parental care:

Uniparental care: In uniparental care, the mechanism of paternity assurance is usually repeated copulation just prior to oviposition. A female may court a male to induce/carry more of her eggs, presumably because male care is a valuable resource that increases her lifetime fecundity.

Biparental care: Biparental care tends to be favored when sexual selection is not intense, and when the adult sex ratio of males to females is not strongly skewed. For two parents to cooperate in caring for the young, the mates must be coordinated with each other as well as with the requirements of the developing young, and the demands of the environment.

Alloparental care: Alloparental care is a seemingly altruistic and reproductively costly behavior. This parenting strategy involves individuals providing care to non-descendent offspring. There are both adaptive benefits and potential costs of alloparenting to the individuals involved.

Maternal care vs Paternal care:

- Maternal care is the most exclusive form of parental care and the most rudimentary form and is provided by females who incorporate toxins into their eggs, oviposit them in protected places or cover their eggs with a hard shell or wax like compound before abandoning them.

- Paternal care is provided by only one parent (usually the male). Males may invest in offspring with nutritional offerings to the female in the form of nuptial gifts of captured prey items or even their own bodies. They may transfer proteins or protective substances in a spermatophore: Male katydids, for example-provide a spermatophore during copulation that may be as much as 40% of their body mass. Ex: Male arctiid moths, *Utetheisa ornatrix* provide a different sort of indirect paternal contribution when they transfer protective pyrrolizidine alkaloids to females during mating. These alkaloids are passed to the eggs, which are then unappealing to predators.

The rise of paternal care can be explained by two hypotheses

- **The Mating Effort hypothesis:** It suggests that males may provide care for offspring in an attempt to increase their own mating opportunities and thus enhance their future reproductive success.

- **The Maternal Relief hypothesis:** It proposes that males provide care to reduce the burdens associated with reproduction for the female, which ultimately generates shorter inter-birth intervals and produces more successful offspring.

Forms of parental care:

1. Trophic egg production: Also known as nurse egg cells, they are non-fertile and rich in nutrients. Ex: Passalidae beetle, if 3rd instar larva stridulates, they are fed by trophic egg.

2. Attending eggs and offspring: Treehopper, *Umbania crossicornus,* mothers tilt their elongated pronotum and fan potential threats, protecting their young until adulthood.

3. Protection of mobile young: Mobile young feeding in open area vulnerable to predators and parasitoids. Females standing nearby, not only threatening potential predators, but also stroking wandering nymphs.

- **Guarding young:** Brazilian sawfly, *Themos olfersii,* continuous guarding their larvae even when they fully sclerotized and starts feeding.

- **Poised to defend:** Assassin bug & Shield bugs continuously watch her emerging nymphs to defend her predator.

- **Egg dumping or brood mixing:** Spatial proximity leads to brood mixing and Alloparental care. Communal breeding, egg dumping, takeover of nests and offspring is seen in case of burying beetle.
- **Formidable mother:** The praying mantis *Oxyphthalmellus* spp. positioned herself at base of a twig to intercept the predators.
- **Blocking entry:** In Myrmicinae colony - the major worker blocks the entrance with its saucer shaped head and receives regurgitated liquid from a minor worker

4. Nest building & burrowing:

- **Hiding the evidence:** Synchronous motions of the front tarsi, rake the sand and producing a characteristic pattern which hide its actual nest. Ex: *Bembix* sand wasp
- **Deceiving enemies:** A bombyliid bee fly, parasitoids of solitary wasp larvae lay several eggs into the fake nest entrance made by the wasp.

5. Brooding behaviour and viviparity: An alternative form of protection is to carry eggs or young either internally or externally.

Communication between parents and offspring: Communication is through the - Tactile communication and Chemical signalling.

Chemical signalling includes

1. By direct internal chemical signals- Hormones
2. By indirect external chemical signals- Pheromones

Cost of Parenting:

- Parents invest lots of resources on their offspring. In most oviparous animals, eggs are loaded with yolk. The female parent contributes nutrients in the yolk. Parents incubate eggs and spend their body heat to maintain optimal temperature.
- Parents spend lots of time and energy on their progeny. It can range from few months to years in some cases. During this time as they are busy with the child, they are unable to mate. Parenting involves guarding the eggs or young ones. To do so, many parents risk their life and face predators.

Benefits of Parenting:

- Parental care increases the chances of offspring survival.
- Increases the offspring fitness and quality.
- Offspring condition is improved.
- It increases the future reproductive success (Meunier *et al.* 2012).

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